

- Q.23 How the fire is extinguished in the aircraft?
 Q.24 Describe inertia starter system of aircraft piston engine?
 Q.25 Describe the accessories of an engine.
 Q.26 Explain the working of a mechanical RTD.
 Q.27 What are the methods to protect engine from fire?
 Q.28 What is the cooling medium in engine system and why?
 Q.29 What are the important features of a starter?
 Q.30 Briefly explain the working of a junction type thermometer.
 Q.31 State five properties of an aviation fuel which are different from automobile fuel.
 Q.32 In a multi-engine aircraft, what is the engine starting sequence followed?
 Q.33 What are the measuring devices used in aircraft reciprocating engines.
 Q.34 Explain the factors affecting the performance of an aircraft reciprocating engine?
 Q.35 Briefly explain the working of liquid engine cooling method.

SECTION-D

- Note:** Long answer type questions. Attempt any two questions out of three questions. (2x10=20)
 Q.36 Differentiate a turbocharger from a super charger (4 points each). List the instruments dedicated to these components.
 Q.37 What types of power augmentation systems are used? In what phase of flight these system are used?
 Q.38 Draw a block diagram showing the various aircraft instruments used in engine and briefly explain the purpose of each.

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4th Semester/ Aircraft Maintenance Subject : Aircraft Reciprocating Engine

Time : 3 Hrs.

M.M. : 100

SECTION-A

Note: Multiple choice questions. All questions are compulsory (10x1=10)

- Q.1 What is the primary function of a carburettor in a reciprocating engine?
 a) To cool the engine
 b) To mix fuel and air
 c) To ignite the fuel-air mixture
 d) To exhaust the burned gases
 Q.2 What type of engine is typically used in smaller general aviation aircraft?
 a) Turbofan b) Turboprop
 c) Reciprocating d) Jet engine
 Q.3 What is the purpose of the magneto in a reciprocating engine?
 a) To cool the engine
 b) To provide electrical power
 c) To ignite the fuel-air mixture
 d) To mix fuel and air

- Q.4 What is the name for the position of the piston when it is at its highest point in the cylinder?
- Top Dead Center (TDC)
 - Bottom Dead Center (BDC)
 - Crankshaft
 - Connecting rod
- Q.5 What does the term “stroke” refer to in a reciprocating engine?
- The diameter of the cylinder
 - The distance the piston travels in the cylinder
 - The angle of the crankshaft
 - The speed of the engine
- Q.6 Which of the following is a major component of the lubrication system in a reciprocating engine?
- Magneto
 - Propeller
 - Oil pump
 - Carburetor
- Q.7 What is the main function of the exhaust system in a reciprocating engine?
- To cool the engine
 - The scavenge the cylinders
 - To remove burned gases from the cylinders
 - To provide power
- Q.8 Which of the following is a common type of cooling system used in aircraft reciprocating engines?
- Water cooling
 - Liquid cooling
 - Air cooling
 - Oil cooling
- Q.9 What is the function of the propeller in a reciprocating engine aircraft?
- To generate thrust
 - The cool the engine
 - To mix fuel and air
 - To ignite the fuel air mixture

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- Q.10 Which of the following is a safety precaution when working with reciprocating engines?
- Never disconnect the magneto ground wire while the engine is running
 - Always ensure the engine is properly grounded before starting
 - Use a wrench that is too large for the job
 - Run the engine without proper lubrication

SECTION-B

Note: Objective/ Completion type questions. All questions are compulsory. (10x1=10)

- Q.11 Define Air screw?
- Q.12 What is the main performance parameter of an engine?
- Q.13 Mixing of petrol and air is done in _____.
- Q.14 How many revolution it take to complete a thermodynamic cycle for a two stroke and four stroke engine.
- Q.15 Name two aviation fuels.
- Q.16 Supercharger increases the _____ of charge?
- Q.17 Thermocouple working principle is _____ effect.
- Q.18 Can a propeller pitch can be changed?
- Q.19 What is a slip ring.
- Q.20 Piston engine works better in low/high ambient temperature?

SECTION-C

Note: Short answer type questions. Attempt any twelve questions out of fifteen questions. (12x5=60)

- Q.21 Compare fixed and variable pitch propellers?
- Q.22 Explain the working of turbocharger.

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